Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/28/2005
PI Name:	Newman, Dava J. Ph.D.		
Project Title:	Microgravity Investigation and Crew Reactions in Flight Experiment		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	08/04/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Deselected in December 2005 (per S. McCollum 12/2006). In flight development phase (not yet manifested)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Metaxas, Dimitri (Rutgers the State University of New Jersey) Wilcox, Karen (Massachusetts Institute of Technology)		
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Task Description:	The Adapt research effort focuses on operational procedures and training to quantify astronaut intravehicular activity (IVA) by developing a modular, kinetic and kinematic capability for the ISS. The collection and evaluation of kinematics (whole-body motion) and dynamics (reacting forces and torques) of astronauts within the ISS will allow for quantification of human motion and performance in weightlessness, gathering fundamental human factors information for design, scientific knowledge in the field of dynamics and motor control, technological assessment of microgravity disturbances, and the design of miniaturized, real-time space systems. The Adapt research effort builds on a strong foundation of successful microgravity experiments, namely, the EDLS (Enhanced Dynamics Load Sensors) flown aboard the Russian Mir space station (1996-1998) and the DLS (Dynamic Load Sensors) flown on Space Shuttle Mission STS-62. In addition NASA ground-based research into sensor technology development and development of algorithms to produce three-dimensional (3-D) kinematics from video images have come to fruition and these efforts culminate in the collaborative Adapt flight experiment. The required technology and hardware capitalize on previous sensor design, fabrication, and testing and can be flight qualified for a fraction of the cost of an initial spaceflight experiment. Four DLS/restraints measure astronaut forces and torques. Two standard ISS video cameras record typical astronaut operations and prescribed IVA motions for 3-D kinematics. Forces and kinematics are combined for dynamic analysis of astronaut motion, exploiting the results of the detailed dynamic modeling effort for the quantitative verification of astronaut IVA performance, induced-loads, and adaptive control strategies for crewmember whole-body motion in microgravity. This comprehensive effort provides an enhanced human-factors approach based on physics-based modeling to identify adaptive performance during long-duration spaceflight, which is critically important for astronaut training as well as providing a spaceflight database to drive countermeasure design. Adaptation strategies and appropriate countermeasures for micro- and partial gravity operations are anticipated as deliverables from the Adapt research program to support future exploration initiatives. The objectives of the MICR0-G (Adapt) flight experiment target NASA's roadmap for future space exploration. As such, the Adapt effort will lead to improved crew training techniques and in-flight countermeasures based upon the observed adaptation of astronauts in altered gravity environments. These techniques and countermeasures will not only familiarize crewmembers with their environment in space, they will also begin the locomotor adaptation prior to exposure to another gravitational environment. A deliverable from the Adapt flight experiment will be a biomechanical model that predicts the locomotor control strategies astronauts develop for a given gravitational environment and that adaptation process used to develop them. This model may include the ratio of muscle/joint usage as a function of time after exposure to the new environment and will be physics-based, accounting for realistic, dynamic human motions. In the process, a kinematic and kinetic capability will be developed for NASA that permits quantitative biomechanical analysis of astronaut full-body motions and performance using typical ISS video cameras, instrumented crew restraints and custom software. The Adapt sensors and video analysis tools will provide the opportunity for countless other studies requiring detailed astronaut biomechanics and dynamics (e.g., exercise, disturbance of the microgravity environment, sensorimotor adaptation and control). The Adapt research program provides key enabling technologies for future exploration programs by developing a quantitative biomechanic model that predicts the locomotor adaptation of astronauts, a suite of kinematic and kinetic sensors, miniature electronics, and bioastronautics testbed capability to assess astronaut performance across different gravitational environments. An accurate biomechanic model of adaptation can be used to develop appropriate pre-flight and in- flight countermeasures that aid the the pre-adaptation of astronauts' control strategies prior to exposure to new gravitational environments, thus improving astronaut performance and reducing the risk of injury. Furthermore, the Adapt research effort will provide a kinematic and kinetic measurement capability for the ISS, enabling future biomechanic studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The hardware, software, and analysis techniques proposed herein are a potentially rich source of information for anyone interested in human factors, medical assessment of human performance, life sciences, rehabilitation, and athletic performance. With further development these prototype systems could be used in clinical and rehabilitation laboratories to assess gait, posture, and locomotion-related diseases. The design of miniature embedded electronics has foreseen Earth Benefits for medical applications by providing much smaller data acquisition systems in clinics, for field research on Earth to supply wearable computing capabilities, and in the home where information technology systems with reduced size and greater capabilities are desired. The adaptive locomotor strategy model could further knowledge in the area of human athletic performance. The kinematic and kinetic measurements suite is also applicable to earth-based athletic performance and competition.
Task Progress:	This past year, the Adapt research program has met several key milestones. The Adapt experiment team successfully completed the Preliminary Design Review (PDR). PDR completion required clear definitions of all Adapt requirements, outlined in the Experiment Document (ED) and the System Requirements Document (SRD). In addition, detailed hardware design drawings were also prepared and presented at the PDR. Based on the PDR drawing package, four prototype Adapt sensors were designed, built and demonstrated at the PDR. Prototype software was also presented at the PDR. Testing is currently underway as the design matures leading up to the Adapt Critical Design Review (CDR). Substantial modeling progress has also been made over the past year. Astronaut models incorporating arm and leg segments have been developed. Algorithms for quantifying the locomotor control motions based on the Adapt kinetic and kinematic measurements have been developed and rigorously tested in simulation. These models apply not only to astronauts in microgravity, but also in partial gravity environments, to support future exploration missions. Furthermore, initial research into neural models for movement adaptation has begun and will continue in the coming year.
Bibliography Type:	Description: (Last Updated: 03/20/2019)
Abstracts for Journals and Proceedings	Ferguson PA, Coleman CP, Newman,DJ. "Characterization of Human Locomotor Control Strategies and Adaptation Across a Spectrum of Gravitational Environments." 55th International Astronautical Congress, Vancouver, Canada, October 2004. 55th International Astronautical Congress, Vancouver, Canada, October 2004. , Oct-2004